



A Comprehensive Review and Analysis of Energy Market Mechanisms and Power System Flexibility



Pilar T. Parada-Mayorga^{1*}, Iván D. Serna-Suárez², Martha L. Torres-Barreto¹, Oscar A. Quiroga Quiroga²

¹ Escuela de Estudios Industriales y Empresariales, Universidad Industrial de Santander, 680002 Bucaramanga, Colombia

² Escuela de Ingenierías Eléctrica, Electrónica y de Telecomunicaciones, Universidad Industrial de Santander, 680002 Bucaramanga, Colombia

* Correspondence: Pilar T. Parada-Mayorga (pilar2117670@correo.uis.edu.co)

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Abstract: The integration of variable renewable energy (VRE) sources has driven research into the flexibility capabilities of power systems, which are characterized by high variability and uncertainty. Flexibility refers to a power system’s ability to respond to changes in demand and generation across different time frames. This concept has been extensively studied in the literature, so the great variety of flexibility definitions and market approaches is a challenge for new stakeholders interested in the field. Establishing a market design that promotes the participation of flexible sources and ensures proper compensation is essential. This paper provides a comprehensive review of market designs proposed in the literature to enhance power system flexibility and approaches for quantifying its economic value. The study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology for the identification, screening, and inclusion of documents, using Web of Science (WOS) and Scopus databases. After analyzing 102 papers, including 50 literature reviews, common approaches and concepts were identified and categorized into demand response, storage, market design, and other general frameworks. Among the market design proposals, the Flexible Ramp Products and Local Flexibility Markets are highlighted, along with an analysis of how to value this flexibility. This study complements existing literature by grouping the most relevant literature on power system flexibility and its valuation in energy markets, clarifying how market designs contribute to addressing renewable integration challenges—essential for future energy system planning with increased renewable energy penetration.

Keywords: Power system flexibility; Market mechanisms; Renewable energy sources; Review; Flexibility value

1 Introduction

The goal of reducing carbon emissions CO₂ has encouraged the growth of generation from renewable sources, which must be integrated into the system in a reliable and sustainable manner [1]. To achieve this goal, power systems must have greater flexibility, since increased levels of variable generation require increased needs for flexibility on power systems, a term that refers to the system’s ability to respond to unexpected changes in demand and supply [1–3], arising from the variability and intermittency that characterizes renewable generation sources.

The improvement of the flexibility of the power system is linked to market mechanisms that can encourage the participation of more flexible sources [3, 4], as well as the improvement of the existing ones, which can offer the required balance, while the penetration of renewables energy grows, regulators must examine rules like curtailment, energy imbalance, ramping and scheduling to balance the interest of consumers, industry and other stakeholders while ensuring a fair remunerations for the market participants and maintain the capacity adequacy to support the transition to low-carbon power system [5].

This study presents a literature review that initially addresses the definitions and classifications proposed to quantify the flexibility of a system. The review was carried out applying the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology, using the Web of Science (WOS) and Scopus databases. A

total of 102 articles were included in the analysis, focusing on studies linking flexibility with market design and high-impact contributions to flexibility research. The results reveal that flexibility has been extensively studied, with 50 literature reviews identified among the examined sample, demonstrating the maturity and scholarly attention this topic has received.

This review aims to provide an accessible entry point for readers looking to understand power system flexibility and its connection to market design, especially for those without extensive prior technical knowledge in this field. It presents various approaches to defining and measuring flexibility and highlights stakeholders' motivations in the valuation of flexibility. This provides a comprehensive starting point for those entering this evolving research area, which is essential for future energy system planning aimed at increasing the integration of renewable energy resources.

Our paper contributes to the relevant literature in the following ways: First, it reviews key documents that present different frameworks for understanding power system flexibility. Second, we compile and examine various market designs proposed to improve power systems' flexibility and discuss their challenges, considering different stakeholders. Finally, we highlight the various definitions of the value of flexibility and introduce a methodology that translates these findings into inputs for the energy planning process, supporting strategic decisions on optimal market design.

2 Methodology

2.1 Contextualization

The integration of renewable energy sources into power systems has been a key strategy to achieve decarbonization goals [4, 6, 7]. Yet, their low cost and dependence on weather conditions that are uncertain and variable, has increased the challenges of integrating this type of resource into current market and system models [8]. This creates the need for sufficient resources to maintain system balance with adequate flexibility, while simultaneously raising the critical question of how to appropriately value and compensate this operational capability.

Many definitions of flexibility exist in the literature, depending on the time scale, or whether it refers to technical properties, systems, or market designs. A widely accepted definition is the following: Flexibility in power systems refers to the ability of system components to adjust their operating points in a timely and coordinated manner to accommodate both expected and unexpected changes in system operating conditions [9]. This property is essential necessary to obtain a large integration of renewable energy sources [10–12].

The main motivation of this review is to understand the state of the art in the study of power system flexibility over recent years, along with market mechanisms designed to promote the participation of flexible generators and the economic quantification of this property.

2.2 Application of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Methodology

We followed the PRISMA guidelines [13] (see Figure 1). The main steps can be summarized as: Identification of Research, Study Selection, Study Quality Assessment, Data Extraction, and Data Synthesis, using the Scopus and WOS databases.

The next section provides details of the screening process conducted through the research data collection, starting from identification to final inclusion in the literature review.

2.2.1 Research questions

This review addresses two fundamental questions in the context of modern power systems transformation. First, it seeks to identify and synthesize the principal studies that examine revisions and consolidations of flexibility concepts in power systems, particularly as they relate to the integration of variable renewable energy (VRE) sources.

Second, it aims to explore the body of research on the necessary adaptations and reforms to energy market structures, mechanisms, and frameworks to effectively address the challenges posed by large-scale renewable energy integration, with particular emphasis on methodologies for assessing, quantifying, and valuing flexibility services in these evolving market environments.

2.2.2 Eligibility criteria

Eligibility criteria focused on two primary dimensions: publication quality and topical relevance. For publication quality, only peer-reviewed journal articles and reviews published in English between 2001 and 2025 with full-text accessibility were included, excluding conference proceedings, book chapters, and editorial materials.

The 2001–2025 timeframe was selected to capture the evolution of flexibility concepts and market design from early renewable integration initiatives through contemporary high-penetration scenarios. For topical relevance, studies were required to address either (1) flexibility in power systems—including conceptual frameworks, measurement approaches, or operational strategies in the context of renewable energy integration—or (2) energy market design, regulatory mechanisms, or flexibility valuation methodologies.

Studies focusing on tangential topics such as supply chain management, isolated material science applications, Power-to-X vehicle technologies without grid integration, circular economy applications without energy market connections, or COVID-19 impacts were systematically excluded.

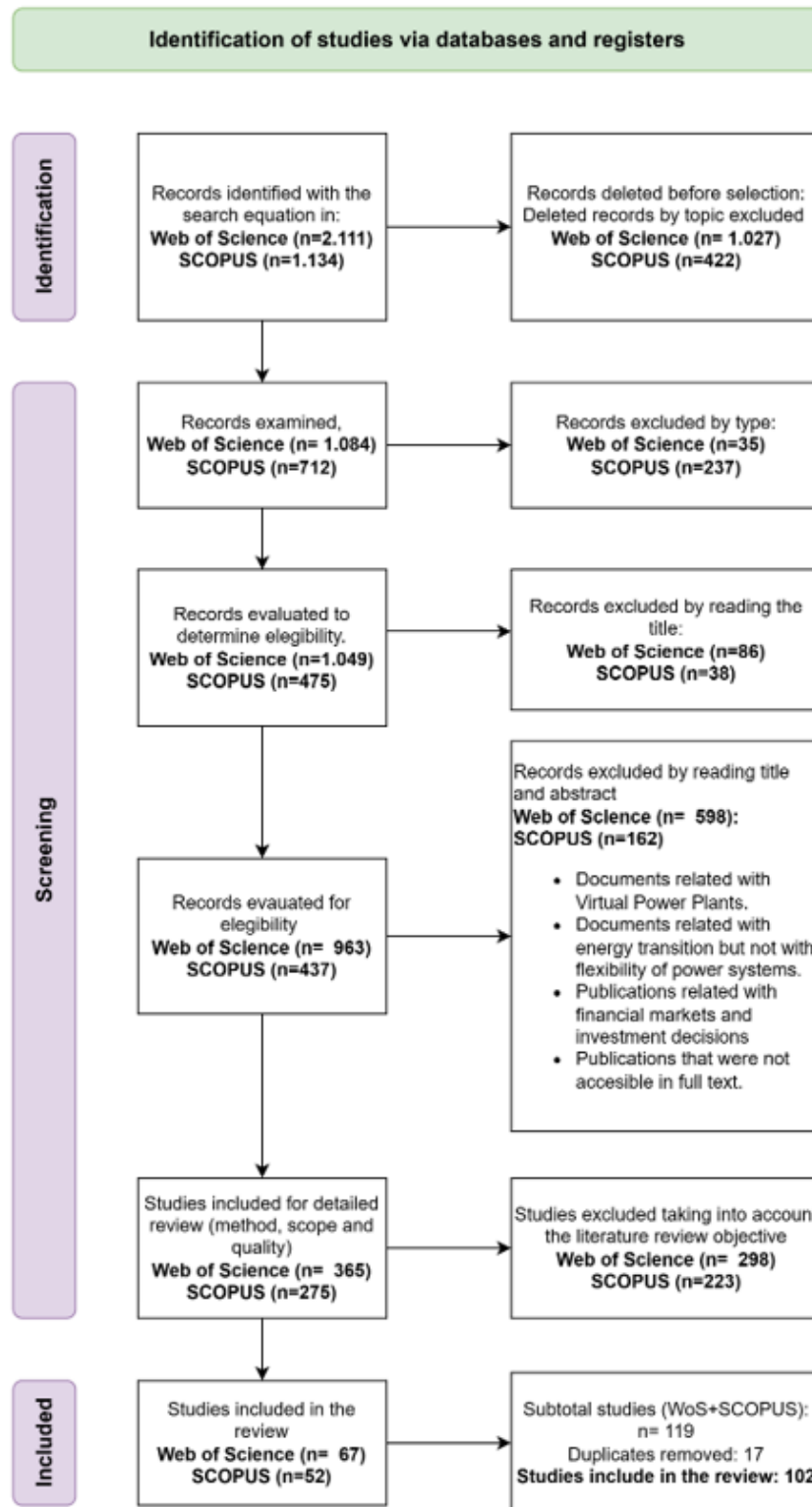


Figure 1. PRISMA flow diagram of study identification, screening, and inclusion process

Note: The diagram illustrates the different stages of the systematic review process following PRISMA guidelines, including identification, screening, eligibility and inclusion. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

2.2.3 Information sources

The review focuses on WOS and Scopus, which offer comprehensive coverage of peer-reviewed literature, including major IEEE publications. While domain-specific repositories like IEEE Xplore contain additional conference proceedings and technical reports, the substantial overlap with Scopus (estimated at 70–90% for IEEE content) and our focus on high-quality, peer-reviewed research justified this scope. Future work could explore grey literature and emerging conference proceedings for complementary insights.

2.2.4 Search strategy

The search equation used in WOS and Scopus included the following terms, and the results were last updated on February 11, 2025, covering research published from 2001 to 2025.

WOS
ALL = ((variable generation OR renewable generation integration OR Uncertainty OR Renewable energy OR penetration OR intermittent renewable generation) AND (flexibility OR Power system flexibility OR Flexibility metrics OR flexib*) AND (Survey OR Review OR literature review OR systematic) AND (electricity markets OR ancillary services OR local flexibility market OR market design OR market clearing OR market mechanisms OR innova* OR market))

Scopus
(TITLE-ABS-KEY (“variable generation” OR “renewable generation integration” OR uncertainty OR “Renewable energy” OR penetration OR “intermittent renewable generation”) AND TITLE-ABS-KEY (flexibility OR “Power system flexibility” OR “Flexibility metrics” OR flexib*) AND TITLE-ABS-KEY (survey OR review OR “literature review” OR systematic) AND TITLE-ABS-KEY (“electricity markets” OR “ancillary services” OR “local flexibility market” OR “market design” OR “market clearing” OR “market mechanisms” OR innova* OR market)).

2.2.5 Selection process

The selection process comprised three mandatory phases—identification, screening, and inclusion—each with explicit, pre-determined criteria to ensure transparency and reproducibility. Database-specific filtering strategies were necessary due to inherent differences in categorization systems between WOS and Scopus.

Table 1 provides a comprehensive breakdown of the WOS screening process, detailing the number of records identified, screened, and included at each decision point, along with specific exclusion criteria and rationale. Table 2 presents the corresponding information for the Scopus database. Both tables document the systematic reduction of records from initial search results ($n = 2,111$ for WOS; $n = 1,134$ for Scopus) to final inclusion ($n = 67$ for WOS; $n = 52$ for Scopus), ensuring full methodological transparency.

For records retrieved from the Scopus database, the following systematic screening process is shown in Table 2.

2.2.6 Record management and de-duplication

The systematic search identified 2,111 records in WOS and 1,134 in Scopus. After screening and eligibility assessment (Figure 1), 67 studies from WOS and 52 from Scopus were selected (subtotal: 119 articles). Duplicate removal eliminated 17 records, resulting in 102 unique articles for final inclusion. Among these, 50 were systematic reviews or state-of-the-art studies, and 52 were core research articles selected for detailed analysis. Full-text versions were obtained from publishers including IEEE, MDPI, and Elsevier for comprehensive review.

Table 1. Identification and analysis via Web of Science (WOS) records

Stage	Process/Source	Details/Criteria
Identification	Records identified through search query WOS ($n = 2,111$)	Records excluded based on WOS category ($n = 1,027$). To identify relevant records, the search was restricted to the following WOS categories: Energy Fuels, Green Sustainable Science Technology, Materials Science Multidisciplinary, Environmental Sciences, Environmental Studies, Management, Engineering Electrical Electronic, Economics, Business, Operation Research Management Science, Engineering Industrial, Engineering Manufacturing, Computer Science Information Systems, Engineering Environmental, Business Finance, Engineering Multidisciplinary, Computer Science Interdisciplinary Applications, Multidisciplinary Sciences, Computer Science Artificial Intelligence, Engineering Mechanical, Mathematics Interdisciplinary Applications, Computer Science Software Engineering, Mathematics.

Continued on next page

Stage	Process/Source	Details/Criteria
Screening	Records screened (<i>n</i> = 1,084)	Records excluded (<i>n</i> = 35): Ineligible document types (proceeding papers, early access, book chapters, editorial material, corrections). Only articles and review articles retained.
	Records assessed by title (<i>n</i> = 1,049)	Records excluded after title screening (<i>n</i> = 86): irrelevant topics including "supply chain digitalization" in companies, "perovskite solar cells", "fabrication of biogas digesters", "electric vehicle charging strategies", "corporate investment", "institutional voids", "distribution system behind the meter DERs", "Information Security Systems", "Optimizing Software", "small molecular nanoparticles", "energy resilience", "Circular Economy of Plastic", "pyrolysis process", "internal combustion engines", "soft wearable electronics", "biological tissue mechanics", "nature conservation in agricultural landscapes", "Public funding", "Microgrids", "residential demand-side management", "machine learning in power-to-X processes", "financial flexibility", "foreign institutional investor", "cryptocurrencies markets", "Geothermal Market", "energy communities", "Biomass gasification", "smart grids".
	Records evaluated for eligibility (<i>n</i> = 963)	Records excluded after abstract screening (<i>n</i> = 598): The off-topic studies including Power-to-X technologies, electric vehicles, financial risk, virtual power plant (VPP) integration in South Africa, lifecycle emissions of electric vehicles (EVs) and fuel cell vehicles, electricity market forecasting, technical VPP aspects, micro gas turbines, materials and prototype development, and green hydrogen systems. Additionally, publications with inaccessible full texts were excluded.
	Studies included in qualitative synthesis (<i>n</i> = 365)	Records excluded for not meeting review objectives (<i>n</i> = 298). Reason: Primary focus on energy markets.
Included		Studies included in review (<i>n</i> = 67) from WOS

This review synthesizes research published between 2011 and 2025, with emphasis on recent developments. We prioritized high-impact publications and foundational works that continue to shape current research directions. Approximately 48% of cited references are from the most recent five years (2021–2025), ensuring coverage of both emerging trends and established knowledge.

2.2.7 Limitations and possible omission risk

Our search was limited to WOS and Scopus. While these databases index a substantial portion of domain-specific content, some specialized conference proceedings and technical reports available exclusively in repositories such as IEEE Xplore may have been omitted. Future research could expand coverage by including direct searches of domain-specific databases to identify additional specialized literature.

2.3 Bibliometric Results

Based on the key co-occurrence analysis shown in Figure 2, four main thematic groups related to market design and flexibility in power systems were identified. These groups can be analyzed considering the four existing drivers of flexibility: storage, smart grid, demand response, and ancillary services.

Flexibility and Storage (Red Cluster): These two concepts are crucial for achieving the energy transition as they ensure sufficient storage resources and require the design of appropriate market mechanisms that incentivize improvements in the services offered within a high renewable energy system.

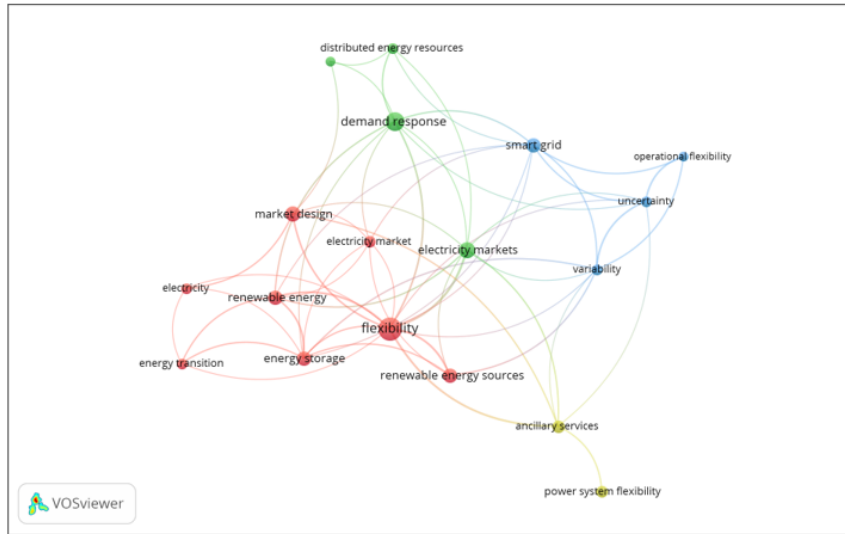
Demand Response and Electricity Markets (Green Cluster): The role of demand has proven to be a key aspect in the design of electricity markets, in particular, distributed energy resources (DER) have grown in the field, reflecting the community's interest in the energy transition.

Flexibility and Smart Grid (Blue Cluster): To deal with the uncertainty and variability of the VRE is necessary to develop smart grid management technologies, and corresponding market designs that provide information and incentives for their participation.

Flexibility and Ancillary Services (Yellow Cluster): The innovation in ancillary services is crucial to increasing the flexibility of power systems and supporting their reliable operation.

Table 2. Identification and analysis via scopus records

Stage	Process/Source	Details/Criteria
Identification	Records identified through search query: Scopus ($n = 1134$)	Records removed before screening ($n = 422$) Reason: Ineligible Scopus subject categories (retained only: Engineering, Energy, Computer Science, Business Management and Accounting, Environmental Science, Mathematics, Decision Sciences, Economics, Econometrics and Finance)
	Records screened ($n = 712$)	Records removed ($n = 237$) Ineligible document types (proceeding papers, early access, book chapters, editorial material, corrections). Only articles and reviews remained.
Screening	Records assessed by title ($n = 475$)	Records excluded based on title screening ($n = 38$) Language: non-English publications excluded Document status: only final articles remained Irrelevant topics: culture/entrepreneurship, energy storage, engineering design/manufacturing, hydrogen systems, smart grids, resilience analytics, project management education, supply chain, biogas, carbon capture, solar collectors, virtual power plants (VPPs), circular economy, COVID-19, investment models
	Records evaluated for eligibility ($n = 437$)	Records excluded after abstract screening ($n = 162$) Irrelevant topic/scope Full text unavailable
	Studies included in qualitative synthesis ($n = 275$)	Records excluded ($n = 223$) Reason: Focus on energy markets; did not meet inclusion criteria
Included		Studies included in review ($n = 52$) from Scopus

**Figure 2.** Keyword co-occurrence network visualization

3 Flexibility Requirements and Metrics

The absence of a unified definition of the term flexibility and the varied use of terminology can lead to misunderstandings, making it difficult for stakeholders to effectively communicate each others [2], this confusion can hinder the transition from mature technology to investment decisions and deployment, affecting the efficiency of the power system operations, so for this reason the definition of flexibility has been a challenge [1, 14, 15].

Flexibility of power systems has become relevant due to the intention of integrating a large number of renewable sources to the system. This property can be classified as flexibility from generation, demand, grid [14], storage, among others, which depends on the technical properties of the assets. The design of energy markets is motivated to encourage the participation of more flexible sources as well as to design adequate mechanisms that efficiently

remunerate all participants. They may include new players, rules, and market products.

The signs of inflexibility [16], include: Inability of the grid to maintain frequency, significant renewable energy curtailments, area balance violations, negative market prices and price volatility. The flexibility is not a new problem, system demand has varied, and generation outages have occurred for as long as power system have existed [17], so this classification can be divided in: Smart grid initiatives, energy storage systems, sectorial integration, new ancillary services, market design improvement, flexible conventional units, demand side flexibility utilization and grid interconnections.

3.1 Quantifying Flexibility Requirements

The flexibility metric can identify the time intervals during which a system is most likely to face a shortage of flexible resources. It can measure the relative impact of changing operational policies and the addition of flexible resources. For example, proposes the Insufficient Ramping Resource Expectation, a metric designed to assess power system flexibility for use in long-term planning [18]. This metric represents the expected number of observations when a power system cannot cope with the predicted or unpredicted changes in net load. Through literature, the next strategies for enhancing flexibility are: advanced energy storage technologies, demand response programs, grid expansion and interconnection, and sophisticated forecasting methods [19].

3.2 Sides of Flexibility

The term flexibility has been studied and classified from different perspectives according to [17], namely: demand side flexibility options, supply side flexibility options (generation-side flexibility), network-side flexibility options, energy storage systems [15] and other sources of flexibility: energy system integration (PtX), energy markets, regulatory policies.

The study of market's flexibility, to the best of our knowledge, began with the work of Ela [9], which is related to the proper market design to incentivize flexibility in both: investment and operation. The first attempt to quantify flexibility indices was made by Makarov [20] who proposed three indices: (1) The ramping limit defined as the maximum change a unit can effect on its operating point in a certain time, (2) power capacity that refers to minimum and maximum power outputs of any generation source and (3) energy capacity related with energy supply of a power source. Later publications introduced ramp-duration as another flexibility index, which denotes the time period that a unit can continue to change its output. Finally regarding demand-side flexibility a new metric called theta is proposed to measure the response time of Demand Response (DR) units [21].

Flexibility can be understood in relation to the elements that make up the system, as well as to its location within the electricity supply chain. This classification includes flexibility associated with generation, grid, storage, and demand [2], since its inception, the literature up to 2012 has analyzed power system flexibility and the consolidation of flexibility requirements in energy systems [18]. Subsequent contribution have addressed new market designs such as flexible ramp products (FRP) [22], the importance of defining and quantifying flexibility from the network side [6], the development of a metric for measuring flexibility from demand side [23] and finally, a historical reviews of definitions of flexibility, sources, evaluation, and parameters [1].

4 Previous Literature Reviews

The growing interest in quantifying power system flexibility, as well as in defining and developing metrics, is reflected in the number of literature reviews published on the subject. Previous reviews can be classified into different categories: demand response, storage, market design, and other holistic approaches emphasizing the importance of the subject and the contributions addressed in the present study (see Table 3).

4.1 Classification Process

Each document was assigned to the category that best represents its main contribution to power system flexibility research. Papers covering multiple themes were classified by their dominant research focus. The classification framework for energy systems review papers was created using the title, abstract, and keywords and the inclusion and exclusion criteria shown in Table 4.

Literature review publications related to flexibility could be grouped into the different sides of flexibility.

A random sample of 15 papers (approximately 30% of the total corpus) was independently coded by a second researcher familiar with power system flexibility literature. The initial inter-rater agreement was 80% (12 of 15 papers), indicating substantial consistency.

For the three papers with initial disagreement, both coders engaged in a structured discussion, reviewing the papers' objectives, methods, results, and conclusions to determine the predominant thematic focus. Consensus was reached for all three cases through this deliberative process, with final classifications based on the primary research contribution as evidenced by the distribution of content across sections and the stated research objectives.

Table 3. Review papers classification

Power System Flexibility Sides Literature Reviews	References
Demand Response	[24–29]
Demand Side Management	[30–35]
Energy Storage	[36–42]
Flexibility Products	[43, 44]
Energy Markets	[45–55]
Flexibility General Reviews	[1–3, 7, 17, 19, 21, 56–66]

Table 4. Inclusion and exclusion criteria by category

Category	Definition and Inclusion Criteria	Exclusion Criteria
DR	Programs and technologies that motivate and empower end-users to modify electricity consumption based on price signals, grid conditions, or system operator commands.	Studies focus on supply-side generation control, physical energy storage systems that do not involve demand modification, and market structures lacking demand response mechanisms.
DSM	Studies on comprehensive strategic planning and execution of activities aimed at influencing customer energy consumption and changing load profiles, including energy efficiency, conservation, and demand response programs, often with long-term planning horizons.	Papers that focus only on supply-side solutions, energy storage technologies, or market trading mechanisms.
Energy Storage	Energy storage technologies and systems in various forms (electrical, thermal, mechanical, chemical), covering technical features, applications, grid integration, and operations strategies.	Papers mainly focus on demand modification without physical storage, generation technologies, market design without storage emphasis, or solely transmission and distribution infrastructure.
Market Design	Electricity market structures and regulatory frameworks, wholesale market design including pricing mechanisms, market-clearing processes, and capacity, energy, and ancillary services markets.	Studies that focus only on technical operations, non-market technologies, DR/DSM without market mechanisms, or non-market coordination.
Flexibility Products	Studies focusing on the design, characterization, standardization, and trading mechanisms of flexibility as a product in energy markets.	Studies not considering product design, general market design without flexible products, papers with marginal flexibility discussions, lack market context.
General Framework	Comprehensive reviews integrating multiple power system components (demand-side, storage, markets), holistic grid strategies, integrated energy systems, or broad frameworks.	Papers focused on a single category, specialized scope, or clear classification within DR, Storage, or Market Design.

Note: DR = Demand Response; DSM = Demand Side Management.

4.2 Demand Response Side Reviews

The transformation of consumers into prosumers who can actively participate in energy markets through onsite production and flexible consumption, adjusting their demand according to market signals, has motivated the study of demand side flexibility [6, 33]. Their participation will play a significant role in integrating large shares of renewable energy, as demand management strategies allow consumers to support system operations while obtaining economic benefits for their flexibility potential [67].

Table 5 synthesizes the principal challenges, assumptions, barriers and mechanism proposed to leverage flexibility from the demand side flexibility.

Table 5. Key findings from demand-side flexibility reviews

No.	Problems	Mechanism	Assumptions	Barriers
1	Peak demand management and system reliability	Interruptible loads and capacity markets	Customers respond rationally to economic signals	Customer acceptance and participation in the programs
2	Customer engagement and participation barriers	Aggregations platforms	Customers respond rationally to economic signals	Customer acceptance and participation in the programs
3	TSO-DSO coordination challenges	TSO-DSO coordination platforms	Aggregators can efficiently coordinate distributed resources	Policy uncertainties and redefinitions of TSO and DSO responsibilities
4	Grid stability issues	Price-based programs	Infrastructure is available or deployable	Infrastructure limitations
5	Flexibility quantification and valuation	Flexibility markets ahead markets scheduling	Demand exhibits price elasticity	Lack of standardization

Note: TSO = Transmission System Operators; DSO = Distribution System Operator.

The opportunities and challenges in demand response as a source of power system flexibility in energy-intensive industries are classified from the owners' perspective [29], focusing on the integration of these mechanisms into electricity markets—primarily in the USA and Germany. This analysis highlights critical aspects such as contingency reserve utilization, capacity contribution, market design, and flexibility [68]. Notably, demand response has significant potential as a vital component of modern electricity markets to enable system flexibility through programs that target various industries like the automotive sector, biodiesel production, polymer manufacturing, steel manufacturing, paper mills, the pharmaceutical industry, and cement production [69, 70] since the industrial flexibility not only could modify the demand profile, but also reduces the operational cost of the power system to avoid investing in fast-run power plants [26].

Additionally, the integration of distributed generators (DG) and demand-side management (DSM) programs raises the need for renewable generators to develop effective bidding strategies [32, 33], like ahead-markets scheduling that includes day-ahead and intra-day markets [68], and other market models that commercialize the flexibility that demand can provide, highlighting the new roles and functionalities that demand will have and the correct mechanisms to ensure market efficiency and a secure network operation [31], the optimal use of diverse and geographically distributed sources, considering the needs of the power system and the individual consumers [24] the role of the demand in real time markets [25], and the requirement of precise modeling techniques to capture the new technologies constraints and the dynamics of future power markets [28].

Furthermore, the role of the consumer must be empowered, making them the focus of the business model and considering their flexibility as essential to achieving system balance, where the DR programs have different timescales ranging from several years to real-time [27]. The residential sector has helped overcome many challenges related to high energy production costs during peak periods, as well as issues of reliability, security, and congestion management in both generation and distribution systems [30]. Incentives for customers to modify their demand could be implemented through a mechanism design that employs direct messaging and centralized decisions [31], which could enhance the welfare of all market participants, ensure truthful self-reporting of customer preferences, and enable the development of optimal pricing techniques while maintaining privacy.

4.3 Storage Side Flexibility Reviews

The challenges in the planning and inclusion of storage technologies in the systems, and how this can enable flexible generation and stable electricity supply required by consumers, have been studied through different systematic literature reviews [36, 37].

The economic valuation of energy storage is highly dependent on operation time, availability of flexibility options and sector coupling [4, 70]. Energy storage systems represent a significant source of system flexibility through ancillary service provision [41], enabling increased renewable energy utilization and addressing future electricity supply-demand challenges [36]. However, despite their potential contribution to sustainable power networks, Energy Storage Systems (ESS) deployment has been constrained by high technology costs, limited implementation experience, and regulatory uncertainties within conventional electricity markets [37]. Accelerating ESS adoption

requires stable regulatory frameworks and market structures that incentivize technology development. Beyond addressing market barriers, technological advancement must focus on improving efficiency, reducing costs, and minimizing environmental impacts across all storage technologies, including mechanical systems [39]. See Table 6.

Energy storage systems offer critical advantages for power systems, notably enhanced supply security and operational flexibility. Realizing these benefits requires appropriate market design frameworks and strategic expansion planning models to guide investment decisions, particularly given accelerating renewable energy integration [38]. To facilitate ESS deployment, various market mechanisms have been implemented, including participation in ancillary services and capacity markets, as well as wholesale market access supported by policy incentives such as feed-in tariffs, capacity payments, and investment tax credits [40]. Determining optimal ESS specifications—including both power and energy capacity requirements—can be accomplished through multiple methodologies, such as analysis of historical wind generation profiles or probabilistic approaches based on wind power forecast error distributions [42].

Table 6. Key findings from storage-side flexibility reviews

No.	Problems	Mechanisms	Assumptions	Barriers
1	Grid stability	Capacity Markets, Spinning and non-spinning reserve	Technology performance improves	Uncertain revenues and business models
2	Market design inadequacies for storage	Ancillary services using: BESS: Battery Energy Storage Systems, ESSs: Energy Storage Systems, TESSs: Thermal Energy Storage Systems, HPs: Heat Pumps	Storage technology costs will continue to decline	Regulatory uncertainty, Lack of standardized valuation methodologies
3	Transmission congestion and infrastructure constraints	Congestion Management	Grid infrastructure can accommodate storage integration	High capital costs, Efficiency losses in charge-discharge cycles
4	High costs and economic viability	Power-to-X and sector coupling, Vehicle-to-grid (V2G)	Regulatory frameworks for support storage	Geographics constrains, Material toxicity and disposal concerns

4.4 Design Markets Flexibility Side Reviews

The pioneering work of Lund et al. [45] establishes critical linkages between electricity market design and power system flexibility, particularly examining how market structures enable flexibility in renewable-intensive systems. An analysis of 400 flexibility-related references demonstrates that substantial flexibility can be achieved through strategic energy management, suggesting that power systems may accommodate high VRE penetration without necessarily requiring transformative infrastructure investments [45, 46]. However, as VRE deployment accelerates, net load patterns will evolve significantly. A key market consequence will be increased demand for balancing services and intraday trading, reflecting the inherent forecast uncertainty of renewable generation in day-ahead timeframes [47].

Building on these foundational insights, Table 7 synthesizes findings from major literature reviews on market design for flexibility, systematically organizing the specific problems addressed, mechanisms proposed, underlying assumptions, and implementation barriers identified across the literature.

Economically efficient flexibility trading among distribution system operators and market participants can be facilitated by Local Flexibility Markets (LFMs)—electricity trading platforms operating in geographically limited areas such as neighborhoods, communities, and small cities [71]. Within LFMs, residential prosumers, DER, and industrial prosumers function as flexibility providers, coordinated by aggregators for market participation. On the demand side, DSOs procure flexibility for operational needs, including voltage control, congestion management, and loss reduction, while balance responsible parties acquire flexibility to optimize portfolios and minimize imbalance costs [48]. While comprehensive flexible market-price systems remain under development, novel business models for local energy trading are emerging through peer-to-peer (P2P) and transactive energy frameworks [52], which necessitate technological infrastructure including smart metering, blockchain-enabled smart contracts, intelligent storage systems, and well-designed market incentive mechanisms [49].

Table 7. Key findings from design markets flexibility side reviews

Topic	Specific Problems	Mechanisms/Solutions	Key Assumptions	Barriers/Challenges
Relationship between market design and flexibility in energy systems [45–47, 51]	Balancing supply-demand with intermittent VRE. Grid stability and technical integration challenges. Impact on wholesale markets and generation assets. Regional Balancing of VRE variability.	Strategic integration of demand, energy, and supply side flexibility. Grid Infrastructure enhancement. Coordinated Market rules and reserve market redesign. Smart grids, microgrids, and regional cooperation mechanisms.	Infrastructure availability. Transmission capacity can be expanded. Technology maturity and cost-effectiveness. Regional Cooperation is feasible and beneficial.	High costs for advanced technologies. Regulatory complexity and frameworks gaps. Transmission constraints. Cost allocation challenges. Behavioral barriers.
Local Flexibility Markets Design [48, 49, 52]	Coordination of flexible resources at the distribution level. Manage local grid constraints.	Standardization of flexibility products. Multi-level market coordination. Definition of centralized or bilateral trading.	Local resources can provide flexibility. The market mechanism can efficiently allocate flexibility. Coordination TSO-DSO is achievable. Prosumers are willing to participate.	Lack of standardization. Small market size and liquidity issues. High transactions costs. Limited flexibility and resource availability.
Local Energy Markets [50, 53–55]	Enable local energy trading. Prosumer participation and engagement. Business Model Viability. Optimal market-clearing process decisions. Microgrids management.	Peer-to-peer (P2P) trading platforms. Blockchain-based trading. Different auction mechanisms and heuristic algorithms.	Technology enables automated trading. Local trading benefits all participants. Information is accessible, and the computational resources are adequate.	Regulatory and legal barriers. Privacy concerns, Lack of standardized platforms. Business model uncertainty. Real-time computation challenges.

Note: TSO = Transmission System Operators; DSO = Distribution System Operator, VRE = variable renewable energy.

In addition to the LFMs, the design of Local Energy Markets (LEMs) has attracted significant research interest in recent years due to increasing DER penetration [50]. Within this framework Gao et al. [54] further categorize bidding models into equilibrium models and agent-based models, providing insight into the underlying algorithmic structures. To transform passive consumers into active market participants, various incentive mechanisms must be implemented, including dynamic market pricing, reduced network fees, innovative pricing structures for members of energy communities or microgrids, direct peer-to-peer trading, and price negotiation between prosumers using game-theoretic strategies [63]. These mechanisms must accommodate diverse forms of aggregation and participation across day-ahead, intraday, balancing, and real-time markets.

5 Market Designs and the Power System Flexibility

The challenge of a market design that considers all the issues related to the VRE integration has been widely studied [46, 51, 72]. There is a consensus about the “right market design for the future”, where the market clearing process should manage the required level of flexibility among the generation resources by ensuring that adequate flexible capacity is available, and send appropriate price signals for resources to continue offering their flexibility [73]. However, “it is unclear whether current market designs are incentivizing resources that have flexible capabilities to offer to the short-term energy and/or ancillary services market when active power flexibility is needed the most”, so the markets designs is still on debate, since it will need to ensure the flexible resources are incentivized to provide their flexibility when needed [51, 74].

The transition to a new market design will require economic principles, like those proposed by Newbery et al.

[75] that suggest policies such as supporting more interconnection and better remuneration of flexibility services, and De Vries and Verzijlbergh [76], who argues that market design requires intertemporal arbitrage, as in the presence of enough local flexibility. The mechanisms could be classified into procuring additional reserves, flexible ramp products, and FRP [77].

The existing and traditional market design elements in a centralized dispatch like: frequency scheduling and short settlement intervals, ancillary service markets, make-whole payments and day-ahead profit assurance payments, have incentivized the suppliers to offer flexibility to the market operator and allow it to commit and schedule the supplier's output [78]. These recent market designs could be: flexibility from nontraditional resources, evolving regulating reserve markets, ancillary service markets for primary frequency control and flexible ramping products [79].

The common factors among the proposals for new market designs can be classified into system flexibility, ancillary services, capacity mechanisms, granular pricing, regulation, and others. See Table 8.

Table 8. Design markets, factors, challenges, and principles

Factors	Challenges and principles	References
System Flexibility	Inclusion of nontraditional resources of flexibility.	[78, 80–82]
	Promotion of operational flexibility.	
	Flexible Ramping Products and regulating reserves.	
Ancillary Services	New markets for frequency control, reserves, and balancing services.	[78, 81, 82]
	Integration of AS with Renewable Energy Sources (RES).	
Capacity Mechanisms	Design of Capacity Markets.	[79, 80, 82, 83]
	Long-term contracts markets.	
	Interoperability with other related markets.	
Granular Price	Increase temporal and spatial granularity of spot prices.	[79, 81, 83]
	Scarcity pricing mechanism.	
	Market power monitoring.	
Regulation and others	Centralized auctions for long-term energy contracts.	[80–84]
	Participation of distributed generation.	
	Model the uncertainty of RES generation through stochastic dispatch.	
	New Market Players.	

According to Zhou et al. [80] the challenges of integrating large amounts of RES in electricity market design include: ensuring revenue sufficiency, providing flexibility and reliability, and market power monitoring and mitigation. This redesign of the markets are based on these principles: Integrity across time, space, and externalities, cost and risk minimization, allocation of risk to those who can best bear it, compensation of fixed costs with fixed income streams and variable costs with variable income streams, flexibility to respond to future events, co-optimization of multiple objectives might provide the capability of lower investment and operating system costs [62, 85].

Likewise, another element in the wholesale market design includes: temporal and spatial granularity of spot prices, cost or bid-based arrangements for dispatch and price formation, scarcity pricing, ancillary services, capacity mechanism, centralized auctions for long-term energy contracts, and the geographical integration of different market segments [72, 81].

The characteristics, benefits, and challenges of the market design models identified are explained in Table 9.

Table 9. Market designs models

Reference	Market Design Name	Market Design Model	Benefits	Challenges
[43, 86, 87]	Flexible Ramp Product (FRP)	Inclusion of a restriction related to the FRP in the Unit Commitment Model.	Adding FRP facilitates wind integration and improves reliability.	The estimation of net load uncertainty and the FRP availability on conventional sources.

Reference	Market Design Name	Market Design Model	Benefits	Challenges
[88]	Ramping Ancillary Service (RAS)	Ramp capacity for future intervals. Ascending and descending ramp.	Ensures the availability of the ascending and descending ramps.	Designed for markets based on audited costs, such as Chile. Increases computational complexity (e.g., every 5 minutes).
[89]	Short Term Energy (STE) Contract	Agents can submit bids for specific amounts of energy every 15 minutes using a predefined price based on the day-ahead market. Markets with nearly real-time transactions to dynamically adjust supply and demand. Resolution of 5 to 15 minutes.	A new bilateral energy contract has been introduced to address the imbalance caused by variable renewable energy (VRE) producers.	Complexity in technological implementation.
[6, 85]	STE		Price signals that incentivize generation and demand responses.	Complexity in managing large amounts of data. Sophisticated forecasting and automation tools.
[85]	High-frequency dispatch	Increase in dispatch frequency. E.g., dispatch every 5 minutes.	Rewards flexible generators. Reduces the need for non-market mechanisms.	High implementation costs and operational complexity. Increased market power of certain agents.
[85]	Guaranteed-price energy auction	Explicitly remunerate flexibility during hours of low net demand through auctioning.	Provides signs of flexibility.	An appropriate auction design that does not generate ineffective results. Not tested in real markets.
[85]	Monitoring Capacity Markets outputs	Monitor whether the capacity market results are sufficiently flexible.	Ensure that the calculated capacity is sufficient to meet flexibility requirements.	Designing appropriate metrics.
[85]	Equivalent Firm Power	Contracts where renewable generators are grouped with conventional ones to guarantee firm power. Separate markets: one for conventional “on-demand” generation and another for renewable generation “when available”.	Simplified market design. New opportunities for storage.	Untested. Ineffective combinations of resources may arise.
[85]	Split Markets		Negotiation of differentiated products.	Not tested.
[62, 85, 90]	Co-optimization Energy and Capacity	Joint optimization of energy and capacity	Fewer re-dispatches and off-market actions.	To find a balanced solution.

Reference	Market Design Name	Market Design Model	Benefits	Challenges
[66, 91]	Capacity Remuneration Mechanisms (CRM)	Ensure that sufficient generation capacity is available to meet demand.	Encourages investment in generation capacity and maintenance of existing capacity.	Increased cost to the end user.

5.1 Flexible Ramp Products

Flexible Ramp Products (FRPs) introduce two new market design variables to existing Unit Commitment (UC) and Economic Dispatch (ED) formulations: Flexible Ramp-Up (FRU) and Flexible Ramp-Down (FRD) capabilities. These products offer multiple operational benefits, including enhanced management of ramping capacity from controllable resources, reduced frequency of power balance violations, decreased deployment of regulation services, lower occurrence of penalty prices due to ramp scarcity, and improved operational reliability and flexibility [43]. Effective FRP integration requires understanding ramp requirement calculations and demand curve construction. While comprehensive validation through realistic case studies remains necessary, FRPs are expected to reduce reliance on demand offsets in real-time dispatch [44].

Although physical flexibility may be sufficient in many systems, contractual availability often constrains its utilization [74]. FRPs address this gap by increasing system flexibility through additional generator ramping capability, improved generation curve imbalance management, and opportunity cost payments compensating resources providing incremental ramp capacity [73]. The market-clearing process must manage required flexibility levels by ensuring adequate flexible capacity availability and transmitting appropriate price signals to incentivize continued flexibility provision [66].

Effective FRP market design requires careful modeling approaches, with stochastic models demonstrating superior performance compared to deterministic models in managing ramp constraints under uncertainty [92]. Successful implementation necessitates comprehensive market mechanisms incorporating: (1) fair payment systems, (2) transparent market processes, (3) technology-independent frameworks accommodating diverse resources, (4) appropriate incentives and assurances, and (5) performance-based evaluation principles ensuring participants are assessed on their ability to deliver flexible services effectively [92]. Table 10 synthesizes the problems addressed, key assumptions, methodologies employed, implementation barriers, and principal contributions of ramping flexibility products identified in the literature.

The implementation of these flexible products varies across different regions and shows more divergence than convergence, demonstrating that the optimal market design depends on renewable penetration levels, grid topology, existing market structures, and regional operational practices.

The CAISO has implemented a three-settlement system (day-ahead, 15-minute, real-time) with explicit flexible ramping products, reflecting California's high renewable penetration and operational complexity; by the other side, MISO has adopted a two-settlement system (day-ahead, real-time) with flexible ramping products, balancing flexibility needs with market simplicity, NYISO remains in the investigation phase, suggesting cautious evaluation of other markets' experiences before committing to a specific design and European and Australian markets are exploring various flexibility services, with different emphasis on frequency response, reserve services, and voltage control reflecting their distinct grid characteristics.

Table 10. Key aspects of flexible ramp products

Market Design	Problem Solved	Market Product Proposed	Key Assumptions	Contributions	Implementation Barriers
Ramp Capability Co-optimization [73]	Ramp shortages causing price volatility and power balance issues with high variable renewable energy (VRE) integration	Ramp capability model co-optimized with energy and ancillary services; system-wide and zonal ramp requirements.	Ramp capability specified for 10-minute response time.	Proposed model reduces scarcity and price volatility, implementable via opportunity costs or separate pricing.	Current practices (increased reserves) create market distortion and balancing costs.

Market Design	Problem Solved	Market Product Proposed	Key Assumptions	Contributions	Implementation Barriers
Flexible Ramping Products [74, 86]	Load-generation imbalances from variable renewable generation (RG); lack of ramping incentives	Flexible ramping products (CAISO FlexiRamp); pay-for-performance reserves. Flexible Ramp Up (FRU) and Flexible Ramp Down (FRD) integrated into RTED and Unit Commitment	VG increases variability requiring faster response; centralized scheduling more efficient; 5-minute settlements better than hourly. FRP requirements function of net load variability/uncertainty.	5-minute settlements more effective than hourly; performance-based payments better incentivize speed/accuracy; nontraditional resources significantly improve efficiency. Significant cost savings from improved uncertainty estimation.	Bilateral contracts and self-scheduling limit flexibility; unclear if current designs sufficient; regional markets haven't converged on single approach. High solar penetration requires high FRP morning/evening; lack of structured pricing.
	Inadequate traditional reserves for sharp load variations; ramping scarcity from forecast errors	Two-settlement (MISO) and three-settlement (CAISO) market clearing systems; new flexible ramping products	High RES penetration requires more operating reserves; conventional plants offer less than technical capability for profit; flexibility needed short-term and long-term	Stochastic models more effective than deterministic with ramp constraints; clarifies challenges across supplier types; identifies measurements for required/available flexibility. Comprehensive categorization of flexibility resources and metrics; clear definition incorporating variability/uncertainty management; higher VRE penetration increases value of flexibility options.	Strategic supplier behavior (offering less than capability); infrastructure needs for demand-side participation; environmental maintenance costs from thermal ramp provision. Large-scale VRE integration jeopardizes security; sole dependency on energy storage may become futile; market ineptitudes compromise flexibility; gaps in mathematical evaluation techniques.
Comprehensive Flexibility Services [66]	Reliable operation; supply-demand imbalances	Comprehensive flexibility products: FRP (FRU/ERD), capacity markets, ancillary services (FCRs, frequency response), reserve services, voltage control, security services	Continued massive VRE integration; flexibility depends on net-load characteristics; must maintain security at reasonable cost across timescales		

The evolution of the market designs approaches for managing ramping flexibility in power systems with high renewable penetration, shown through first the explicit flexible ramping products with separate pricing mechanisms (such as CAISO's FlexiRamp and MISO's flexible ramping proposals), to enhanced uncertainty modeling using advanced statistical techniques and integration of ramping products with capacity markets and ancillary services.

5.2 Flexibility Markets, Local Electricity Markets, and other Business Models in the Energy Markets

The integration of distributed renewable resources into traditional grid structure, while enabling active consumer participation and maintaining system reliability has motivated the design of different novel market structure, that

can be categorized in: Local Electricity Markets, which show how to structure energy market organizations and the technical market mechanism and algorithmic solutions for local trading; Emerging Business Models that explain how to organize structures and actor relationship in decentralized systems and the viability of new business models, and finally the LFM, that describe the characteristics and analyze the operational implementation of a flexibility products market, allowing the trading of flexibility supplied by both producing and consuming units at the distribution level [71], where the key design challenges of these markets are: TSO-DSO coordination, coexistence of different flexibility services, the prosumer [93].

This section synthesizes research across three complementary market categories: LEM, Emerging Business Models, and Local Markets for Electricity Trading (LFM). See Table 11.

Table 11. Principal findings from Local Energy Markets (LEM), business models, and electricity trading research

Category	Problem Solved	Market Product Proposed	Key Assumptions	Contributions	Implementation barriers
LEM [50, 53, 94]	The integration of distributed energy resources poses barriers to consumer participation. The transition from centralized to decentralized system management	Community-based markets with decentralized direct trading, with active consumer participation	Smart grids that enable real-time information. Cost competitiveness required. Local Market Operators platforms	Can lower costs through flexibility. Incentivize remuneration from flexibility. Profit opportunities through strategic coordination	Lack of specific legal frameworks. High complexity to solve the uncertainty related to the integration of distributed generation. High implementation costs.
Emerging Business Models in LEM [52, 59, 60, 93]	Need for enterprise models to increase the integration of renewable sources generation through the procurement of flexibility.	Business models proposed for prosumers, retailers, prosumers in the peer market, pure consumers, pure generators, storage operators, platform operators, aggregators, representatives, and grid operators.	Customer-centric approach. Digital technologies as key enablers (AI, IoT, blockchain). Prosumers trade flexibility as commodity. Decentralization improves flexibility and resilience	Access to different markets can diversify revenues. Improvement of flexibility provision	Lack of peer-to-peer trading laws. Revenue/cost structures missing that must contribute the investment decisions. Large-scale implementation and blockchain energy consumption.
Local Markets for Electricity Trading (LFM) [44, 63, 71]	Disaggregated small/medium customer flexibility that cannot access markets. Traditional centralized systems are inadequate for decentralized distributed generation management	Markets where products like ramping capacity and capacity services are traded	Smart metering infrastructure is available. Regulatory evolution toward market liberalization. Prosumers become active service providers	Benefit from resources through flexibility services. Enhanced flexibility for RES variability and better real-time balance	Aggregator roles are undefined. Lack of standardized flexibility product definitions. TSO-DSO coordination complexity.

6 The Value of Flexible Generation and Flexibility

The key value definitions in the literature are related to two terms: VRE Value and Value of Flexibility. These terms are analyzed in the next section.

6.1 Value Definitions

Having a clear definition of “Value” is essential because it acts as the starting point for determining how to assign economic worth to an asset or service, such as flexibility. The literature review presents various definitions, so more analysis is needed to develop a definition accepted by both the academic and industrial sectors.

The definition of “Value of flexibility” has been subject to various operationalizations in the literature, with authors adopting different approaches based on their analysis of the market and system. These conceptual frameworks include: cost perspective, value perspective, immediacy value, ramping capability value, cost savings, and average realized power prices. See Figure 3.

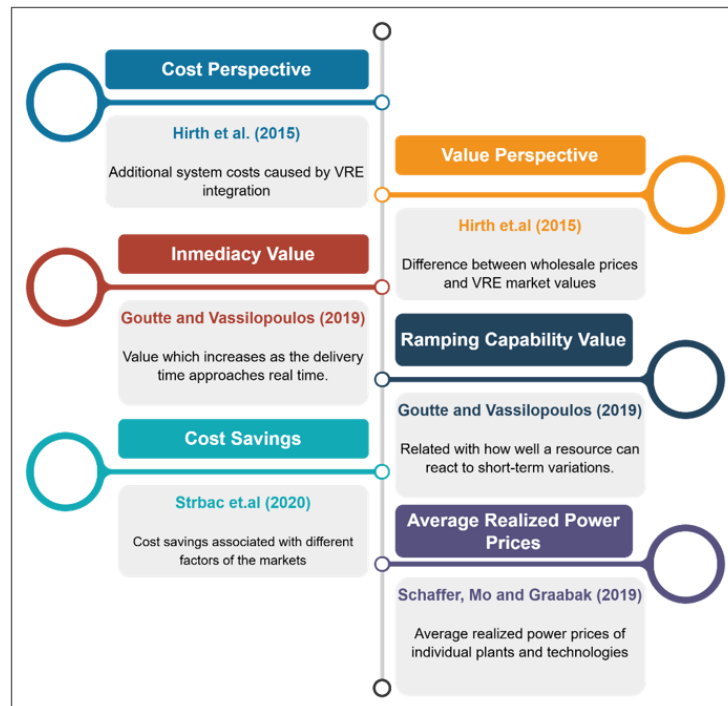


Figure 3. Conceptual frameworks to define the value of flexibility

Trying to find a definition is important to consider the consumers’ viewpoint of a new energy market, who are primarily motivated by financial benefits—reduced electricity bills—and environmental concerns when considering participation in demand flexibility programs [95], and the use of a proper mathematical formulation that helps the planner simulate and model the power system, which could improve its security and reliability and make the power system more economical [96] taking into account metrics of flexibility and the analysis and framework used to quantify VRE integration costs, as explained by Hirth et al. [97].

One of the VRE market value definitions is the revenue that generators can earn on markets, without income from subsidies [98], alternatively, Hirth et al. [97] show that the integration of VRE costs can be estimated from a cost perspective, as the additional system costs caused by VRE integration; or from a value perspective, as the difference between wholesale prices and VRE market values, defining the value factor as the ratio between a technology’s market value (unit revenues) and wholesale electricity prices. This ratio gives an insight into how much revenue a VRE technology can generate compared to the prevailing market prices for electricity, e.g., a value factor of 1 (or 100%) indicates that the unit revenues received by VRE generators are equal to the average wholesale price. In contrast, a value factor less than 1 suggests that VRE unit revenues are falling below average wholesale prices.

Another approach to define value, is to measure of how flexible resources can respond to short-term price variations in day-ahead and intraday markets, named the immediacy value which increases as the delivery time approaches real-time, highlighting the urgency and associated risks in the intraday markets; and the ramping capability which relates how well a resource can react to short-term variations, particularly in 15-minute intervals compared to hourly intervals [99].

In contrast to the above, another definition of the value of flexibility [82] is related to the potential to realize significant cost savings associated with: avoidance of energy curtailment from low-carbon generation sources, efficient provision of operating reserve and response-related balancing services, potential savings in generation capacity, and avoidance of network reinforcement/addition. The savings are obtained from the reduction in system capacity requirement—low carbon generation, conventional generation, transmission, interconnection, distribution assets—and lower operating cost (due to energy curtailment avoidance, CO₂ cost savings, and reduced fuel usage).

Alternatively, another definition is proposed by the study [100], which assesses the value of flexibility through the average realized power prices of individual power plants and technologies. This realized power price is the average price per unit of energy that a given power station or technology achieves over a defined time period. The performance of a plant or technology refers to the percentage difference in the realized power price for a plant or technology compared to the average power price in the same area. The performance is used as a measure of how flexible a plant or technology is in adjusting generation to variations in the power price.

As explained, the variability of power prices increases with the penetration of ERV, which enables flexible technologies, such as storage and demand response, to operate effectively and profitably. In this scenario, the assessment of the incremental cost for transitioning into a flexible system must be taken very carefully, because if the models underestimate price variability, it could lead to inadequate investment in technologies, thereby affecting the flexibility of the power system [101]. Additionally, a highly flexible system requires significant infrastructure investment, which can substantially raise the cost of power supply [102].

Besides the need for a globally accepted definition of the value of flexibility, new modeling and decision-making tools are also needed that can capture the complex interdependencies of a RES-based system [103]. Since even with pricing schemes that internalize flexibility costs, it is not sufficient to make flexible units profitable, additional changes to market design are required [104], and in this way, encourage financing mechanisms for investment in renewable energy sources [105].

6.2 Stakeholder Perspective of Flexibility Value

Flexibility value in power systems is inherently multi-dimensional, varying by stakeholder perspective, temporal scale, and decision context. Table 12 provides a comparative framework organizing six distinct flexibility value assessment approaches according to their definition, primary stakeholder, relevant time horizon, measured attribute, and applicable decision scenario.

Table 12. Comparative framework for flexibility value assessment definitions

Definition	Stakeholder	Time Scale	What It Measures	When It Applies
Cost Perspective	System planner/Regulator	Long-term planning (years to decades)	Total economic cost of VRE, including integration costs added to generation costs.	Investment decisions, policy evaluation, and comparing technologies' total system impact
Value Perspective (Market Value)	Generator/Investor	Medium to long-term (months to years)	Revenue potential measured as wind/solar-weighted average price relative to baseload price.	Investment feasibility, plant profitability assessment, optimal VRE deployment
Immediacy Value	Balancing Service Provider	Very short-term (minutes to hours before delivery)	Premium captured by responding close to real-time, revealed in intraday vs. day-ahead price spreads.	Real-time trading, forecast error compensation, balancing market participation
Ramping Capability	System operator/Flexible generator	Operational short-term (15-min to hourly)	Technical ability to adjust output rapidly, valued through sub-hourly product premiums.	Frequency regulation, sub-hourly markets (15-min vs. 60-min products), managing renewable ramps
Cost Savings	System planner/Society	Short to long-term (operational to planning)	Avoided costs in generation capacity, networks, and operations from deploying flexibility.	System adequacy planning, infrastructure investment deferral, welfare optimization
Average Realized Power Prices	Generator	Operational (hours to seasons)	Actual revenue per MWh achieved considering when generation occurs.	Performance benchmarking, operational optimization, measuring flexibility premium vs. baseload

The flexibility valuation methods span from decades-long planning horizons to minute-by-minute operations. Cost and value perspectives serve long-term decisions: planners assess total VRE integration costs for policy evaluation, while investors measure revenue potential for project feasibility. At operational scales, immediacy value quantifies real-time response premiums through intraday-day-ahead price spreads, while ramping capability captures sub-hourly adjustment value for frequency regulation. Cost savings measure avoided infrastructure and operational expenditures for adequacy planning, while average realized prices provide operational performance benchmarks for generators.

This taxonomy demonstrates that no single metric captures flexibility value comprehensively; rather, appropriate valuation requires matching the analytical approach to the stakeholder and decision context.

6.3 Recent Methodological Advances (2025)

Real Options theory has emerged as a powerful framework for quantifying flexibility value in power systems by extending financial option pricing principles to physical assets and operational decisions [106].

The Real Options framework provides a unifying analytical lens for understanding flexibility across multiple dimensions of modern energy systems. Operational flexibility—embodied in dispatchable generation, energy storage, demand response programs, and advanced grid technologies—creates economic value precisely through its capacity to respond optimally to uncertain market conditions, variable renewable generation, and shifting demand patterns [106]. Similarly, strategic flexibility in investment decisions allows system planners to defer, expand, or abandon projects as uncertainties resolve over time, avoiding costly irreversible commitments.

This work addresses an important gap by showing how methods for valuing energy derivatives and assessing smart grid technology align, both based on optionality principles essential for decarbonizing and decentralizing energy systems.

7 Integrated Energy Planning and Management Approach

The frameworks discussed above should be transformed into practical tools and market mechanisms through a planning process that includes, first, an asset classification step; second, a quantification process that may incorporate the flexibility metrics suggested in the literature; and third, an integration step that accounts for new flexibility products and market designs. The goal is to identify actions needed to ensure a safe and reliable energy supply as the integration of VRE sources grows.

This approach is proposed to provide a possible route for policymakers' analysis, investment decision information, and to offer a general view of the possibilities based on the reviewed literature, considering how different value definitions could influence decisions today.

Each part of the approach includes assumptions, methods, and outputs that help develop medium- and long-term plans, as well as questions that guide the application of these stages. The best market design depends on renewable penetration levels, grid topology, existing market structures, and regional operational practices. See Figure 4.

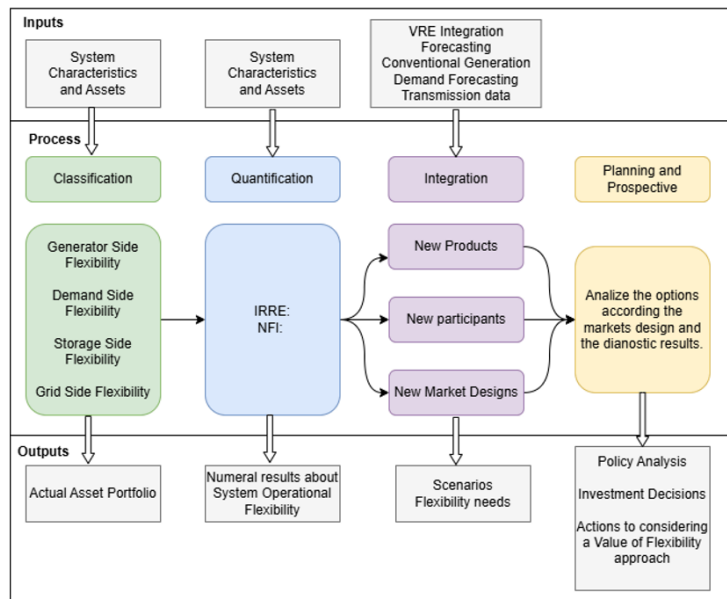


Figure 4. A methodological framework for flexibility assessment and market integration in energy system planning

In each of the Planning Process proposed the next question should be solved:

- *Classification Process*: What types of flexibility exist in the energy system and how they can be categorized?

The Classification stage aims to systematically identify, categorize, and characterize all available flexibility resources within the energy system based on their technical and operational aspects attributes. The primary objective is to create a structured inventory of flexibility assets that serves as the basis for subsequent quantification and valuation, ensuring that all potential sources and its characteristics are properly identified.

- *Quantification Process*: How can we quantify the technical flexibility of different assets?

The Quantification stage transforms the qualitative asset portfolio from classification into quantitative flexibility indicators that enable objective comparison across different technology types and system contexts, presented through comprehensive datasets, performance indicators, and valuation metrics.

- *Integration Process*: How can quantified flexibility be effectively incorporated into energy markets and operational practices?

The Integration stage aims to design and choose the market mechanisms, regulatory frameworks, and operational procedures that enable effective participation of flexibility resources in energy markets and system operations, using the outputs data of the quantification process and the forecasted data of VRE generation, demand consumption forecast and technical and transmission data, to decide which elements of the three options for improve flexibility, such as new market products, new market participants and new market designs, could be considered for the planning process.

- *Planning and Prospects Process*: How should flexibility considerations be integrated into models for integrated resource planning and transmission expansion planning?

The Planning stage outputs describe policy analysis reports and investment decision frameworks that guide strategic pathways for energy system stakeholders. These results empower decision-makers to develop adaptive, evidence-based energy strategies that leverage flexibility to navigate the energy transition efficiently and equitably.

8 Outstanding Questions

Implementing the market designs proposed in the literature—whether new market products, new participant categories, or redesigned energy market structure requires coordinated action among all electricity supply chain stakeholders. However, critical knowledge gaps still exist that fundamentally hinder both practical deployment and systematic comparison across studies. The next gaps are still a challenge due to specific barrier to turning theoretical market designs into operational:

- Current flexibility valuation methods assume deterministic or well-characterized stochastic processes, failing to capture the option value created by deep uncertainty in future VRE penetration. This gap creates a barrier of implementation since without real options frameworks that account for irreversible investments, system planners cannot justify flexibility investments to regulators, and investors face unquantifiable risk premiums that can undervalue flexibility relative to conventional generation, leading to suboptimal capacity mix. Critical research priorities include developing hybrid Real Options Analysis frameworks validated against actual market outcomes, creating accessible practitioner tools, and reconciling long-term strategic valuations with short-term operational assessments.

- While DER can provide flexibility at both distribution and transmission levels, optimal allocation and pricing market mechanisms remain unresolved, with existing designs lacking operational protocols for coordinating procurement between TSOs and DSOs. This creates implementation barriers—TSOs and DSOs may compete for the same resources, creating conflicting dispatch signals and price distortions and DER aggregators face regulatory uncertainty about market access and exchange data protocols that can underutilize available flexibility. Critical research priorities include developing coordination protocols that prevent system-level conflicts, establishing standardized data exchange and settlement procedures, and piloting mechanisms in real-world networks with rigorous performance monitoring.

- Energy planning frameworks must support long-term capacity expansion decisions that align with actual market structures, incorporating flexibility as a core planning element with explicit links to market design to evaluate whether flexibility targets are achievable under realistic conditions. This requires standardized planning frameworks that accommodate different assumptions, technical characteristics, and policy objectives. Critical research priorities include developing integrated frameworks that endogenize market design impacts on flexibility provision, validating planning models against empirical data from mature flexibility markets, and creating decision-support tools that link long-term planning targets with operational market parameters to ensure adequate flexibility in increasingly complex power systems.

9 Conclusions

This review has summarized the evolving landscape of market designs for power system flexibility and the methods for measuring its economic value—a crucial necessity as VRE adoption accelerates worldwide. The literature reveals convergence toward several core design principles: innovative ancillary service products, capacity

markets, enhanced temporal and spatial price granularity, improved forecasting systems, advanced dispatch models including co-optimization and stochastic frameworks.

However, translating these theoretical constructs into operational reality requires a systematic implementation pathway. A three-stage planning process is proposed to bridge this gap: (1) asset classification to identify flexibility sources across generation, storage, demand-side resources, and grid infrastructure; (2) quantification using standardized flexibility metrics that capture ramping capability, response time, and duration; and (3) integration through new market products and regulatory frameworks. This structured approach provides policymakers with actionable guidance for flexibility planning, offers investors clarity on value drivers for flexible assets, and enables system operators to maintain reliability as energy systems transition toward decarbonization.

The concept of flexibility lacks a single, unified definition due to its multi-dimensional nature and its service to various stakeholders across different time scales. Various approaches—such as cost analysis for planning, market value for investments, premiums for immediacy in balancing, cost savings for infrastructure, and realized prices for benchmarking—are valid within specific contexts. However, this fragmentation complicates system-wide optimization. Although real options theory offers a promising framework for valuing flexibility under uncertainty, it encompasses diverse methodologies that require a deep understanding of the underlying assets and rely on strong assumptions about future behavior.

Authors Contributions

Conceptualization, P.T.P.-M., I.D.S.-S., M.L.T.-B., and O.A.Q.Q.; methodology, P.T.P.-M., I.D.S.-S., and M.L.T.-B.; validation, M.L.T.-B. and O.A.Q.Q.; formal analysis, P.T.P.-M. and I.D.S.-S.; investigation, P.T.P.-M. and O.A.Q.Q.; writing—original draft preparation, P.T.P.-M., M.L.T.-B., and I.D.S.-S.; writing—review and editing, M.L.T.-B., I.D.S.-S., and O.A.Q.Q.; visualization, P.T.P.-M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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Nomenclature

AS	Ancillary Services
DAM	Day Ahead Market
DER	Distributed Energy Resources
DG	Distributed Generation
DR	Demand Response
DSM	Demand Side Management
ED	Economic Dispatch
EM	Electricity Market
ESS	Energy Storage Systems
EV	Electric Vehicles
FRP	Flexibility Ramp Products

IRRE	Insufficient Ramping Resource Expectation
ISO	Independent System Operator
LEM	Local Energy Market
LFM	Local Flexibility Market
LMP	Local Marginal Price
P2P	Peer-to-Peer
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PSF	Power System Flexibility
PV	Photovoltaic
RES	Renewable Energy Source
RTM	Real Time Markets
TSO	Transmission System Operators
UC	Unit Commitment
